

Project Two Template

MAT-350: Applied Linear Algebra

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Problem 1

Use the `svd()` function in MATLAB to compute A_1 , the **rank-1 approximation of A** . Clearly state what A_1 is, rounded to 4 decimal places. Also, **compute** the root-mean square error (RMSE) between A and A_1 .

Solution:

```
A = [1 2 3; 3 3 4; 5 6 7];  
[U, S, V] = svd(A);  
  
A1 = S(1,1) * U(:,1) * V(:,1)';  
A1_rounded = round(A1, 4)
```

```
A1_rounded = 3×3  
    1.7039    2.0313    2.4935  
    2.7243    3.2477    3.9867  
    4.9087    5.8517    7.1832
```

```
RMSE1 = sqrt(mean((A - A1).^2, 'all'))
```

```
RMSE1 =  
    0.3257
```

Problem 2

Use the `svd()` function in MATLAB to compute A_2 , the **rank-2 approximation of A** . Clearly state what A_2 is, rounded to 4 decimal places. Also, **compute** the root-mean square error (RMSE) between A and A_2 . Which approximation is better, A_1 or A_2 ? Explain.

Solution:

```
A2 = A1 + S(2,2) * U(:,2) * V(:,2)';  
A2_rounded = round(A2, 4)
```

```
A2_rounded = 3×3  
    0.9878    2.0324    2.9820  
    2.9065    3.2474    3.8624  
    5.0561    5.8515    7.0826
```

```
RMSE2 = sqrt(mean((A - A2).^2, 'all'))
```

```
RMSE2 =  
0.1166
```

Explain: A2 is closer to A than A1. Its RMSE (about 0.12) is much smaller than the rank-1 error (about 0.33) because the second singular term adds back another layer of structure. Each extra singular pair you keep lowers RMSE; a full rank-3 reconstruction would match A exactly.

Problem 3

For the 3×3 matrix A , the singular value decomposition is $A = USV'$ where $U = [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3]$. Use MATLAB to **compute** the dot product $d_1 = \text{dot}(\mathbf{u}_1, \mathbf{u}_2)$.

Also, use MATLAB to **compute** the cross product $\mathbf{c} = \text{cross}(\mathbf{u}_1, \mathbf{u}_2)$ and dot product $d_2 = \text{dot}(\mathbf{c}, \mathbf{u}_3)$. Clearly state the values for each of these computations. Do these values make sense? **Explain.**

Solution:

```
u1 = U(:,1);  
u2 = U(:,2);  
u3 = U(:,3);  
  
d1 = dot(u1, u2)
```

```
d1 =  
1.6653e-16
```

```
c = cross(u1, u2)
```

```
c = 3x1  
-0.1114  
-0.8520  
0.5115
```

```
d2 = dot(c, u3)
```

```
d2 =  
1.0000
```

Explain: These results match what we expect from an orthogonal factor U . Distinct columns of U are orthonormal, so $\mathbf{u}_1$ and $\mathbf{u}_2$ are perpendicular and d_1 is effectively zero. Since U is a proper rotation ($\det(U) = 1$), the cross product $\mathbf{u}_1 \times \mathbf{u}_2$ aligns with $\mathbf{u}_3$, which is why $\mathbf{c}$ matches $\mathbf{u}_3$ numerically and $d_2 = \text{dot}(\mathbf{c}, \mathbf{u}_3)$ is 1.

Problem 4

Using the matrix $U = [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3]$, determine whether or not the columns of U span $\mathbb{R}^3$. Explain your approach.

Solution:

```
rank(U)
```

```
ans =  
3
```

```
det(U)
```

```
ans =  
1.0000
```

Explain:

Problem 5

Use the MATLAB `imshow()` function to load and display the image A stored in the `image.mat` file, available in the Project Two Supported Materials area in Brightspace. For the loaded image, **derive the value of k** that will result in a compression ratio of $CR \approx 2$. For this value of k , **construct the rank- k approximation of the image**.

Solution:

```
load('image.mat');  
A = double(A);  
imshow(uint8(A));  
title('Original Image');
```

Original Image



```
[m, n] = size(A);  
[U, S, V] = svd(A);  
  
CR_target = 2;  
k5 = round((m*n) / (CR_target*(m+n+1)))
```

```
k5 =  
801
```

```
Ak5 = U(:,1:k5) * S(1:k5,1:k5) * V(:,1:k5)';  
imshow(uint8(Ak5));  
title(sprintf('Rank-%d Approximation (CR ~ %d)', k5, CR_target));
```

Rank-801 Approximation (CR ~ 2)



Explain: $\text{rank}(U) = 3$ and $\det(U) = 1$ show that all three columns of U are linearly independent. Three independent vectors in $\mathbb{R}^3$ form a basis, so the columns of U span all of $\mathbb{R}^3$. That is guaranteed by the SVD: U is an orthogonal matrix, so its columns are an orthonormal basis.

Problem 6

Display the image and compute the root mean square error (RMSE) between the approximation and the original image. Make sure to include a copy of the approximate image in your report.

Solution:

```
RMSE_k5 = sqrt(mean((A - Ak5).^2, 'all'))
```

```
RMSE_k5 =  
3.1539
```

Problem 7

Repeat Problems 5 and 6 for $CR \approx 10$, $CR \approx 25$, and $CR \approx 75$. **Explain** what trends you observe in the image approximation as CR increases and provide your recommendation for the best CR based on your observations. Make sure to include a copy of the approximate images in your report.

Solution:

```
CR_list = [10, 25, 75];
k_list = zeros(size(CR_list));
RMSE_list = zeros(size(CR_list));

for i = 1:length(CR_list)
    CR_target = CR_list(i);
    k = round((m*n) / (CR_target*(m+n+1)));
    k_list(i) = k;

    Ak = U(:,1:k) * S(1:k,1:k) * V(:,1:k)';
    RMSE_list(i) = sqrt(mean((A - Ak).^2, 'all'));

    figure;
    imshow(uint8(Ak));
    title(sprintf('Rank-%d Approximation (CR ~ %d)', k, CR_target));
end
```



Rank-64 Approximation (CR ~ 25)



Rank-21 Approximation (CR ~ 75)



k_list

```
k_list = 1×3
      160    64    21
```

```
RMSE_list
```

```
RMSE_list = 1×3
      8.2118   12.3039   18.2656
```

Explain: For an m-by-n image, a rank-k SVD approximation stores k left singular vectors, k singular values, and k right singular vectors. Solving $CR = (m*n)/(k*(m+n+1))$ for k with $CR = 2$ gives $k = 801$ for this image.